

Trainings ▾

General Information

This section outlines the recommended maintenance practices for conventional, hybrid, and organic acid technology (OAT) coolant types.

Maintenance intervals and coolant test limitations are specific to the coolant formulation. This section outlines the different types of coolants and the recommended maintenance practices for coolants.

Coolant Products for Cummins Engines

Table 1 below contains differences between conventional, hybrid, and OAT coolants. Cummins Inc. endorses and recommends Fleetguard® coolants. Table 1 describes some Fleetguard® products in each additive technology category.

Table 1 - Coolant Additive Technology Overview			
Classification	Conventional	Hybrid	OAT
Life:	Short	Medium	Long
Maintenance Requirements:	High	Medium	Low
Product Characteristics:	Heavy Duty coolant. Requires use of supplemental coolant additives (SCA) to replenish depleted additives.	Capable of an extended service interval when suggested maintenance practices are followed. Require use of SCA to replenish depleted additives.	Capable of an extended service interval without addition of SCA or extender products when suggested maintenance practices are followed.

Table 1 - Coolant Additive Technology Overview

Classification	Conventional		Hybrid		OAT	
	Product:	** Color:	Product:	** Color:	Product:	** Color:
*Cummins Recommended Products:	Fleetcool™	Green	ES Compleat™	Blue	ES Compleat™ OAT	Red
	Compleat™	Blue	Fleetcool EX™	Fuchsia	Fleetcool™ OAT	Pink
			PG Plus™	Royal Blue	PG Platinum™	Dark Red
Performance Standards:	Coolants used in Cummins products must meet appropriate Cummins Engineering Standard (CES) for each product. Refer to product Owners manual to confirm the coolant performance CESs required. Most Fleetguard® coolants meet American Society for Testing and Materials (ASTM) D-3306, ASTM D-6210, Cummins® Engineering Standard (CES) 14603 and CES 14439 at a minimum.					

Consult Atmus™ Filtration Technical Assistance for detailed Fleetguard® product information and assistance at 800-223-4583 or www.atmusfiltration.com.

* Product availability varies by region. Visit www.atmusfiltration.com or contact your local customer assistance for more information.

Note : ** Though coolant colors are listed in the table above, color is **not** a reliable method for identifying coolant. Coolant mixing, contamination, and degradation can influence coolant colors. Additionally, other brands and formulations of coolant may have colors very similar to Fleetguard® products and may **not** be compatible.

Topping Off and Dilution of Cooling Systems

Topping off the cooling system with an appropriate pre-diluted or 50/50 coolant is critical for maintaining additive concentrations and freeze protection. **Never** top off cooling system with water **only**. This can lead to deposit formation and inadequate corrosion and cylinder liner protection.

Top off the cooling system using the same coolant formulation used to fill the cooling system. Fleetguard® coolants are compatible with other coolant formulations with similar additive packages, but mixing coolant formulations such that one or more coolant(s) is present in a concentration greater than 25 percent by volume relative to the "original" coolant could result in critical additive depletion, unpredictable performance, and may result in serious engine damage. In the case of switching from one coolant formulation to another, drain the cooling system completely, flush and refill with the desired coolant.

Contact the coolant manufacturer for specific guidance if coolants **must** be mixed.

Coolant Testing

Table 2 contains recommended test intervals for coolants used in Cummins products. See the Routine Maintenance Schedule in Owners Manuals for product specific coolant service interval requirements. Coolant service interval requirements in Owners manuals supersede the recommendations in Table 2.

Table 2 - Test Intervals for Coolants

Test	Conventional	Hybrid	OAT
Glycol ¹	Twice per year minimum	Twice per year minimum	Twice per year minimum
SCA Level ¹	Every lubricating oil drain	Shortest of: 241,401 km [150,000 miles], 4000 hours, or 1 year	Not required
Coolant Replacement Limits Check	Shortest of: 241,401 km [150,000 miles], 4000 hours, or 1 year	Shortest of: 241,401 km [150,000 miles], 4000 hours, or 1 year	Shortest of: 482,803 km [300,000 miles], 6000 hours, or 1 year

1. Additive and glycol testing are also recommended when the following occur:

- Coolant loss between test intervals exceeds 10 percent of system capacity
- Water pump seal, radiator core, or other external leakage
- Replacement or repair of a cooling system component.

Refer to Procedure 379-007 in Section 5 (</qs3/pubsys2/xml/en/procedures/00/00-379-007.html>) for testing methods. Refer to Procedure 379-008 in Section 8 (</qs3/pubsys2/xml/en/procedures/00/00-379-008.html>) for interpretation of test results.

Maintenance Records

Maintenance programs should be accompanied by accurate record-keeping practices. Records **must** be capable of supplying the information required for the following:

- Support diagnostic and troubleshooting procedures involving cooling system
- Support investigation of potentially warrantable malfunctions
- Forecast repairs leading to prevention of malfunctions.

Routine cooling system maintenance records **must** include the following information:

- Date of service and actions taken during service

- Accumulated vehicle and coolant hours
- SCA level, unless using OAT/Extended Life Coolant
- Freezing point or antifreeze concentration as a percent of coolant volume
- Coolant top-off quantity
- Laboratory analysis readings, where available.

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